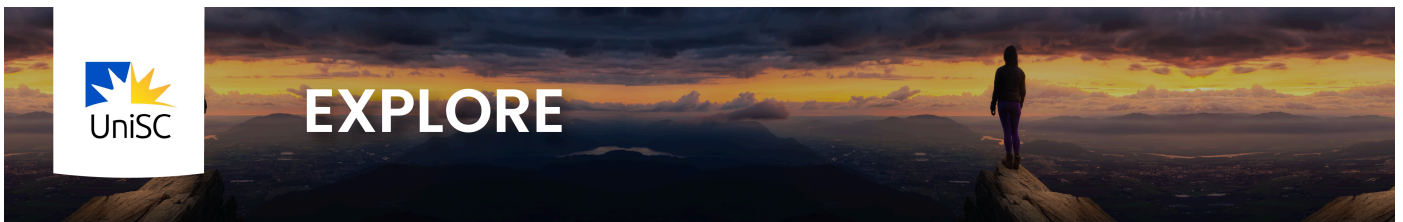


4.2 How is the CAPM used in calculating the cost of capital?



Guiding Questions

1. What is the weighted average cost of capital (WACC)?
2. What is the formula for WACC?
3. Where exactly is the CAPM involved in WACC?
4. How is the cost of debt calculated?
5. What are the important considerations associated with the WACC calculation?



What will you learn?

- How various costs are captured in WACC.
- How CAPM facilitates the WACC calculation.
- How WACC is calculated in practice.
- How firms use WACC.
- How WACC matters in choosing whether to invest in a project or not.

What is WACC?

WACC is short for the **weighted average cost of capital**. The sources of capital in a business typically include debt and various types of equity such as common stock & preferred shares. WACC is calculated by figuring out the cost of each type of capital, weighting these costs by their respective percentages of total capital, and then adding them together.

In finance, WACC is very important as it is the discount rate used in the net present value of a project calculation. In other words, it is the discount rate used in valuing a business. Equivalently, WACC can also be thought of as the required return an investor needs in order to consider investing in a project.

The following are the typical steps involved in calculating WACC:

Step One: Calculate the cost of equity. This is typically done by applying CAPM

Step Two: Calculate the after-tax cost of debt (as interest is a tax deductible expense, it offers a tax shield advantage).

Step Three: Multiply the cost of each type (equity and debt) with its weight and add them together.

What is the formula for WACC?

$$WACC = \left\{ \left(\frac{E}{V} \right) \times R_e \right\} + \left\{ \left(\frac{D}{V} \right) \times R_d \times (1 - T) \right\}$$

where:

E is the market value of the firm's equity (it is the total equity market cap).

D is the market value of the firm's debt

V is the total market value, which is the sum of E and D.

E/V is the weight of equity capital

D/V is the weight of debt capital

R_e is the cost of equity

R_d is the cost of debt.

T is the tax rate.

Solved Example

A company has the following characteristics:

- Market value of equity (E) = \$8,000,000
- Market value of debt (D) = \$2,000,000

- Cost of equity (R_e), from CAPM = 12%
- Cost of debt (R_d), pre-tax = 5%
- Tax rate (T) = 30%

Total value and weights

$$V = E + D = 8,000,000 + 2,000,000 = \$10,000,000$$

$$E/V = 8,000,000 / 10,000,000 = 0.80 \text{ (80\% equity)}$$

$$D/V = 2,000,000 / 10,000,000 = 0.20 \text{ (20\% debt)}$$

Apply the WACC formula

$$\text{WACC} = (E/V \times R_e) + (D/V \times R_d \times (1 - T))$$

$$\text{WACC} = (0.80 \times 12\%) + (0.20 \times 5\% \times (1 - 0.30))$$

$$\text{WACC} = 9.6\% + (0.20 \times 5\% \times 0.70)$$

$$\text{WACC} = 9.6\% + 0.7\%$$

$$\text{WACC} = \mathbf{10.3\%}$$

Interpretation: This company would need to earn at least 10.3% on a new project to create value, since that's the blended cost of the capital it uses to fund it. Notice equity dominates the result (9.6 of the 10.3 percentage points) simply because the firm is 80% equity-financed; the tax shield on debt (multiplying R_d by $(1 - T)$) is what keeps the debt component's contribution relatively small.

Where exactly is the CAPM involvement in WACC?

CAPM is used in calculating the cost of equity:

$$R_e = R_F + \beta(R_M - R_F)$$

How is the cost of debt calculated?

Before we can define the cost of debt, it helps to step back and ask what a bond actually is.

What is a bond?

When a company needs to borrow a large sum of money, it can issue a bond instead of taking a bank loan, essentially an IOU sold to many investors at once. In return for lending the company money, each bondholder receives two things: regular interest payments called coupons, paid on a fixed schedule, and repayment of the original loan amount, the face value, in one lump sum at maturity.

For example, a company might issue a bond with a \$1,000 face value, a 10-year maturity, and a 5% annual **coupon rate**. That means the company borrows \$1,000 today, pays \$50 a year in interest for 10 years, and repays the \$1,000 at the end. The 5% coupon rate is fixed at issuance and printed on the bond; it never changes for the life of that bond.

But bonds are traded, and their price is not fixed. Once issued, a bond can be bought and sold many times before maturity, on a secondary market, just like a stock. The price it trades for changes constantly, because it depends on what investors could currently earn elsewhere on similarly risky debt.

Suppose interest rates in the economy rise after our 5% bond is issued, and new, similar bonds are now issued at 7%. Nobody would pay full price for a bond only paying 5% when 7% is available elsewhere. So, the market marks the old bond down, say to \$850, until the total return a buyer gets at that lower price works out to roughly 7%, matching the going rate.

This is the key idea: **the coupon rate tells you what the company promised to pay when the bond was first issued. The market price tells you what the company would have to pay if it borrowed again today.** These only match on the day of issuance; after that, they drift apart as interest rates move.

Cost of debt is the yield to maturity (YTM) on the company's debt:

$$R_d = YTM$$

We use YTM rather than the coupon rate because WACC is asking, "what would it cost this company to borrow money *right now*?" — and the coupon rate, frozen at issuance, can't answer that. YTM, derived from the bond's current price, tells you exactly what return the market is demanding from this company today. Price and yield always move in opposite directions. A falling bond price is the market's way of demanding a higher effective return from that borrower.

YTM is the discount rate which makes the present value of cash flows associated with a debt instrument equal to its price today. Finance professionals use a finance calculator like the one available [here](https://www.calculator.net/finance-calculator.html)  <https://www.calculator.net/finance-calculator.html> to figure out YTM.

Bond trades...	Relationship
Below face value (a discount)	YTM > coupon rate
At face value (at par)	YTM = coupon rate
Above face value (a premium)	YTM < coupon rate

Calculating YTM: The general formula

$$P = \frac{C}{(1+YTM)^1} + \frac{C}{(1+YTM)^2} + \cdots + \frac{C}{(1+YTM)^n} + \frac{F}{(1+YTM)^n}$$

where P = price, C = coupon payment, F = face value, n = years to maturity.

Worked Example: Calculating YTM

Let's use the bond from earlier: face value \$1,000, 10 years remaining, 5% annual coupon (\$50/year), currently trading at **\$850**.

The problem: find the discount rate r that makes the present value of all future cash flows equal to \$850.

$$850 = \frac{50}{(1+r)^1} + \frac{50}{(1+r)^2} + \dots + \frac{50}{(1+r)^{10}} + \frac{1000}{(1+r)^{10}}$$

There are 10 coupon payments of \$50 each, plus the \$1,000 face value returned in year 10. Unlike the discount bond in Q1 of the practice set (a single cash flow), this equation has 11 separate cash flows, so it can't be solved with simple algebra. This is exactly why finance professionals use a financial calculator or software rather than solving it by hand.

YTM $\approx$ 7.15%

This means that even though this bond's coupon rate is fixed at 5%, an investor who buys it today at \$850 is actually earning a 7.15% annual return. That's the number we'd use as R_d in a WACC calculation for this company. It reflects what the market currently demands to lend to them, not the outdated 5% rate stamped on the bond a decade ago.

Now you try

Consider a coupon bond that has a \$900 par value and a coupon rate of 6%. The bond is currently selling for \$860.15 and has two years to maturity. What is the bond's yield to maturity?

Answer: ~8.50%.

Do you get this answer?

WACC: Various Considerations

For a hypothetical situation that illustrates various considerations associated with the WACC calculation, see the material available at this [link](#) 

(<https://pressbooks.pub/tourocorporatefinance/chapter/3-13-solutions-to-wacc-problems-2/>)_.

WACC: The Case of Queensland Water Businesses

To see a practical example of WACC (calculation for Queensland water businesses), consult the report available [here](https://learn.usc.edu.au/courses/35359/files/3048562?wrap=1) (<https://learn.usc.edu.au/courses/35359/files/3048562?wrap=1>)_.

Solve

A company has:

- Market value of equity = \$15,000,000
- Market value of debt = \$5,000,000
- Cost of equity = 14%
- Cost of debt (pre-tax) = 7%
- Tax rate = 25%

Calculate the company's WACC.

Answer available [here \(https://learn.usc.edu.au/courses/35359/files/3292828?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3292828?wrap=1).

Critically evaluate

1. Suppose an analyst uses book value weights rather than market value weights in calculating the weights of debt and equity in the WACC calculation. Explain why this is problematic.
2. Suppose an analyst treats WACC as a single, constant discount rate throughout the lifespan of a project for valuation. Explain why this is problematic.